APM52009 – Machine Learning for Scientific Computing and Numerical Analysis

Week 4: Introduction to PDEs and Classical Numerical Methods

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Abstract

Partial differential equations (PDEs) model complex phenomena in physics, engineering, and other fields, describing changes in quantities over space and time. Since analytical solutions are rarely possible for real-world problems, numerical methods become essential. This chapter reviews two popular techniques for solving PDEs—finite differences and finite elements. We will cover the algorithmic fundamentals of each method, along with theoretical insight into accuracy, stability, and convergence—providing the tools needed to solve a wide range of PDEs computationally. These numerical solvers will be useful for benchmarking neural network solvers, and creating training data.

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1 Classification of PDEs

Notation We write

$$u_t = \frac{\partial u}{\partial t}, \qquad u_x = \frac{\partial u}{\partial x}, \qquad u_{xx} = \frac{\partial^2 u}{\partial x^2}, \qquad \Delta u = u_{xx} + u_{yy}.$$

Why classify PDEs? Different partial differential equations encode different physical principles—diffusion, wave propagation, steady-state balance, nonlinear transport—and these differences dictate:

- the expected smoothness and behaviour of solutions;
- which numerical schemes are appropriate;
- what stability and convergence guarantees we can rely on;
- why classical PDE solvers are still indispensable in the era of neural networks.

A clear classification is therefore the first step towards understanding which algorithms will work, and why.

Canonical linear models Three fundamental PDEs serve as the prototypes for numerical analysis:

(Poisson)	$-\Delta u = f,$	elliptic: no time, global coupling,
(Heat)	$u_t = \Delta u,$	parabolic: infinite propagation speed,
(Wave)	$u_{tt} = \Delta u,$	hyperbolic: finite propagation speed.

In one spatial dimension these reduce to $-u'' = f$, $u_t = u_{xx}$, and $u_{tt} = u_{xx}$.

Nonlinear models Nonlinearities lead to richer behaviour:

(Allen–Cahn)	$u_t = u_{xx} + u - u^3,$	nonlinear parabolic,
(Kuramoto–Sivashinsky)	$u_t = -uu_x - u_{xx} + u_{xxxx},$	nonlinear parabolic,
(Burgers)	$u_t = -uu_x,$	nonlinear hyperbolic,
(KdV)	$u_t = -uu_x - u_{xxx},$	nonlinear hyperbolic.

Second-order PDE classification For a general operator

$$Lu = au_{x_1x_1} + bu_{x_1x_2} + cu_{x_2x_2} + \cdots,$$

the discriminant $\delta = b^2 - 4ac$ determines the type:

$$\delta < 0 \Longrightarrow \text{elliptic}, \qquad \delta = 0 \Longrightarrow \text{parabolic}, \qquad \delta > 0 \Longrightarrow \text{hyperbolic}.$$

Each class requires different numerical tools; understanding this will be essential throughout the course.

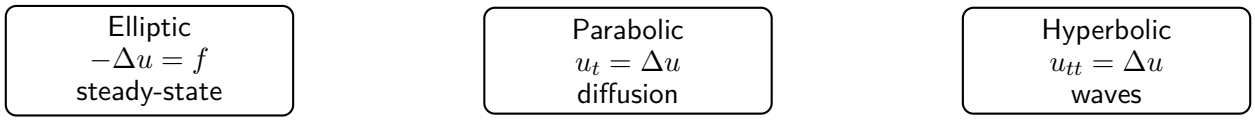

Figure 1: Three fundamental classes of PDEs and their canonical examples.

2 Review of classical methods

Classical numerical methods remain the backbone of scientific computing. They predate machine learning by decades, but they provide something neural PDE solvers do *not*:

- rigorous stability guarantees (energy estimates, CFL conditions);
- provable convergence (e.g., Lax equivalence, Céa lemma);
- certified accuracy and error bounds;
- robustness across geometries and scales;
- reliable **baseline solutions** used to train, benchmark, or validate neural solvers.

Neural networks can approximate PDE solutions, but they do not replace classical methods; instead, they coexist with them, often relying on classical solvers for training data, physical constraints, or safety checks.

Finite differences

FD methods are simple, efficient, and remain standard for fluid dynamics, geophysical flows, and time-dependent problems on simple geometries.

- discretize the domain on a grid;
- approximate the solution locally using low-degree polynomials;
- directly discretize the PDE;
- produce large, sparse linear systems with algebraic accuracy.

Finite elements

Finite elements are the workhorse for engineering applications, structural mechanics, electromagnetics, and multiphysics problems. They:

- discretize the domain with a mesh;
- approximate the solution locally using piecewise polynomials;
- transform the PDE into a variational formulation;
- produce large, sparse linear systems with algebraic accuracy.

Spectral methods

Spectral methods often serve as high-accuracy "ground truth" solvers for validating neural approaches.

- discretize the domain on a grid;
- approximate the solution globally using smooth basis functions;
- directly discretize the PDE, often in coefficient space (e.g., Fourier);
- produce small, dense linear systems with exponential accuracy.

	Finite differences	Finite elements	Spectral methods
discretization	grid	mesh	grid
Geometry	simple	complex	simple
Approximation	local	local	global
Approach	PDE	variational formulation	PDE
Linear system	large sparse	large sparse	small dense
Accuracy	algebraic	algebraic	exponential
Theory	easy (Taylor series)	hard (functional analysis)	medium (complex analysis)
Implementation	easy	hard (mesh)	medium (advanced NLA)
Software	MATLAB pdepe	FEniCS & FreeFEM	Chebfun & Dedalus
Books	[8, 10, 12]	[1, 2, 3]	[6, 13]
At Polytechnique	APM 41012	APM 42031 & APM 52184	–
Golden age	1940–1960	1960–1980	1970–1990
Pioneers	von Neumann & Richardson	Babuška & Courant	Gottlieb & Orszag
In France	Lions & Maday	Ciarlet & Lions	Maday

Historical context and the evolution toward neural PDE solvers

The numerical analysis of PDEs is a remarkably mature field. Between the 1940s and the 1990s, classical schemes such as finite differences, finite elements, and spectral methods were developed and refined to the point where they now underpin almost all industrial and scientific simulations. This "first golden age" established the pillars that remain fundamental today:

- rigorous concepts of consistency, stability, and convergence (Lax);
- variational formulations and approximation theory (Courant, Céa);
- mesh generation, adaptivity, and efficient sparse/dense linear solvers;
- spectral analysis for high-accuracy simulation.

The 2010s brought a new wave of interest: *machine learning for PDEs*. Neural networks, once confined to pattern recognition tasks, began to be used to approximate solutions of differential equations (PINNs), to learn solution operators (DeepONets, Fourier Neural Operators), or to emulate entire simulation pipelines.

Neural approaches are powerful, especially for:

- surrogate modelling on large parametric families,
- reduced-order modelling of complex dynamics,
- learning closures or unknown constitutive laws,
- hybridizing data and physics for ill-posed inverse problems.

But they do *not* replace classical solvers. Modern neural PDE methods depend on them for:

- generating reliable training data;
- computing residuals and numerical losses;
- benchmarking and verification;
- ensuring safety in engineering applications.

This course will emphasize this synergy: neural networks extend what classical methods can do, but they stand on the shoulders of the numerical analysis developed over the past eighty years.

3 Discretization of the 1D Poisson equation

We consider the 1D Poisson equation on $(-1, 1)$

$$-u''(x) = f(x), \qquad u(-1) = u(1) = 0. \tag{3.1}$$

We present two standard discretization schemes: (i) finite differences, and (ii) finite elements.

3.1 Finite differences

Finite difference operators Let $u : \mathbb{R} \to \mathbb{R}$ be smooth and $h > 0$. Taylor expansion around x yields

$$u(x \pm h) = u(x) \pm hu'(x) + \tfrac{h^2}{2}u''(x) \pm \tfrac{h^3}{6}u^{(3)}(x) + \tfrac{h^4}{24}u^{(4)}(\xi_\pm),$$

for some $\xi_\pm$ between x and $x \pm h$. From this we obtain the following.

Definition 3.1 (Finite difference operators)**.**

$$\begin{aligned}
\text{Forward for } u'\text{:} \quad D^h_{x+}u(x) &:= \frac{u(x+h) - u(x)}{h} &&= u'(x) + \tfrac{h}{2}u''(x) + \mathcal{O}(h^2),\\
\text{Backward for } u'\text{:} \quad D^h_{x-}u(x) &:= \frac{u(x) - u(x-h)}{h} &&= u'(x) - \tfrac{h}{2}u''(x) + \mathcal{O}(h^2),\\
\text{Central for } u'\text{:} \quad D^h_{x}u(x) &:= \frac{u(x+h) - u(x-h)}{2h} &&= u'(x) + \mathcal{O}(h^2),\\
\text{Central for } u''\text{:} \quad D^h_{xx}u(x) &:= \frac{u(x+h) - 2u(x) + u(x-h)}{h^2} &&= u''(x) + \mathcal{O}(h^2).
\end{aligned}$$

Definition 3.2 (Consistency and truncation error)**.** A finite difference operator $\mathcal{L}_h$ is *consistent* with a differential operator $\mathcal{L}$ if $(\mathcal{L}_h u)(x) = (\mathcal{L}u)(x) + \tau_h(x)$ with $|\tau_h| \to 0$ as $h \to 0$. The term τ_h is the *local truncation error*; its order ($\mathcal{O}(h)$, $\mathcal{O}(h^2)$, ...) is the *accuracy*.

Proposition 3.3 (Consistency of D^h_{xx})**.** *Let $u \in C^4$ in a neighborhood of $x \in \mathbb{R}$. Then*

$$D^h_{xx}u(x) = \frac{u(x+h) - 2u(x) + u(x-h)}{h^2} = u''(x) + \frac{h^2}{12}u^{(4)}(\xi) \quad \textit{for some } \xi \in (x-h, x+h).$$

In particular, $D^h_{xx}u(x)$ is consistent with $u''(x)$ and has truncation error $\mathcal{O}(h^2)$ as $h \to 0$.

Proof

By Taylor's theorem with Lagrange remainder, for some $\xi_+ \in (x, x+h)$ and $\xi_- \in (x-h, x)$,

$$u(x \pm h) = u(x) \pm hu'(x) + \frac{h^2}{2}u''(x) \pm \frac{h^3}{6}u^{(3)}(x) + \frac{h^4}{24}u^{(4)}(\xi_\pm).$$

Adding the $+$ and $-$ expansions, subtracting $2u(x)$ and then dividing by h^2 gives

$$\frac{u(x+h) - 2u(x) + u(x-h)}{h^2} = u''(x) + \frac{h^2}{24}(u^{(4)}(\xi_+) + u^{(4)}(\xi_-)).$$

By the intermediate value theorem, some $\xi \in (x-h, x+h)$ satisfies

$$\frac{1}{2}(u^{(4)}(\xi_+) + u^{(4)}(\xi_-)) = u^{(4)}(\xi), \quad \text{so that} \quad \frac{u(x+h) - 2u(x) + u(x-h)}{h^2} = u''(x) + \frac{h^2}{12}u^{(4)}(\xi).$$

Therefore

$$\left|D^h_{xx}u(x) - u''(x)\right| \le \frac{h^2}{12}\max_{|y-x|\le h}|u^{(4)}(y)| = \mathcal{O}(h^2),$$

which proves consistency with second-order accuracy.

Remark 3.4. Central differences are second-order for u' and u'', while forward/backward differences are first-order for u'. Higher-order formulas arise from longer Taylor stencils.

Application to 1D Poisson Let $x_i = -1+ih$ with $h = 2/(m+1)$ and unknowns $u_i \approx u(x_i)$ for $i = 1, \dots, m$. Approximating $u''(x_i)$ with the central difference gives

$$-\frac{u_{i-1} - 2u_i + u_{i+1}}{h^2} = f(x_i), \qquad i = 1, \dots, m,$$

where $u_0 = u_{m+1} = 0$ are known from the Dirichlet boundary conditions.

Proposition 3.5 (Tridiagonal linear system)**.** *Collecting the m interior equations yields*

$$A\,\boldsymbol{u} = \boldsymbol{f}, \qquad A = \frac{1}{h^2}\,\mathrm{tridiag}(-1,\,2,\,-1) \in \mathbb{R}^{m\times m}, \qquad \boldsymbol{f} = \{f(x_i)\}_{i=1}^m.$$

For $m = 5$, the system has the form

$$\frac{1}{h^2}\begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \\ f_5 \end{pmatrix}.$$

Remark 3.6. For non-homogeneous Dirichlet conditions, the first and last entries of $\boldsymbol{f}$ must be adjusted to account for the boundary values.

3.2 Finite elements

In this section we introduce the finite element method for the same simple 1D model problem and then place it into an abstract Hilbert-space framework. This will later allow us to reuse the same ideas for higher dimensions and more general operators.

Introduction with 1D Poisson To derive the so-called variational formulation, we multiply Eq. (3.1) by a smooth test function v with $v(-1) = v(1) = 0$ and integrate:

$$\int_{-1}^{1} (-u'')\,v = \int_{-1}^{1} fv.$$

Integrating by parts and using the boundary conditions yields

$$\int_{-1}^{1} u'v' = \int_{-1}^{1} fv.$$

This motivates the bilinear and linear forms

$$a(u,v) := \int_{-1}^{1} u'v', \qquad f(v) := \int_{-1}^{1} fv. \tag{3.2}$$

The variational formulation reads: find u such that

$$a(u,v) = f(v) \qquad \forall v \text{ with } v(-1) = v(1) = 0. \tag{3.3}$$

Note that this formulation requires less regularity than the strong one: it is enough to have $u' \in L^2([-1,1])$.

We then discretize the interval $[-1,1]$ with a uniform mesh. For a given integer $m \geq 1$, define

$$x_j = -1 + jh, \qquad h = \frac{2}{m+1}, \qquad 0 \leq j \leq m+1.$$

We then introduce the standard piecewise linear (hat) functions $\{\phi_j\}_{j=1}^m$ associated with $x_1, \dots, x_m$:

$$\phi_j(x) = \phi\left(\frac{x - x_j}{h}\right), \qquad \phi(x) = 1 - |x|, \qquad \mathrm{supp}(\phi) = [-1,1]. \tag{3.4}$$

These basis functions satisfy the nodal property

$$\phi_j(x_k) = \delta_{jk}, \qquad 1 \le j, k \le m. \tag{3.5}$$

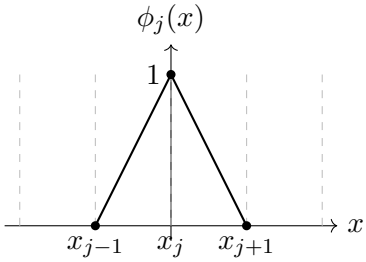

Figure 2: Hat basis function ϕ_j with support $[x_{j-1}, x_{j+1}]$.

We define the finite element space

$$V_h := \operatorname{span}\{\phi_1, \dots, \phi_m\}. \tag{3.6}$$

Any $u_h \in V_h$ can be written as

$$u_h(x) = \sum_{j=1}^{m} u_j \phi_j(x), \tag{3.7}$$

where the coefficients u_j coincide with the nodal values $u_h(x_j)$.

We impose that u_h satisfies the weak formulation (3.3) for all test functions in V_h, i.e., for all hat functions ϕ_i:

$$a(u_h, \phi_i) = f(\phi_i), \qquad i = 1, \dots, m.$$

Substituting $u_h = \sum_{j=1}^{m} u_j \phi_j$ gives the $m \times m$ linear system

$$AU = F, \tag{3.8}$$

with $U = (u_1, \dots, u_m)^\top$, where

$$A_{ij} = a(\phi_j, \phi_i) = \int_{-1}^{1} \phi_j'(x)\, \phi_i'(x)\, dx, \qquad F_i = f(\phi_i) = \int_{-1}^{1} f(x)\, \phi_i(x)\, dx. \tag{3.9}$$

A direct computation shows that A has a tridiagonal structure:

$$A = \frac{1}{h} \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}, \qquad \lambda_j(A) = \frac{4}{h} \sin^2\left(\frac{j\pi h}{4}\right) > 0, \qquad j = 1, \dots, m. \tag{3.10}$$

where $(\lambda_j(A))_{j=1}^{m}$ are the eigenvalues of A. Thus A is symmetric positive definite (SPD), and the discrete problem has a unique solution. Once U is known, we obtain the nodal values $u_h(x_j)$ and the approximation at any $x \in [-1, 1]$ via Eq. (3.7).

Takeaways

- **Finite differences** approximate derivatives using local Taylor expansions, leading to pointwise formulas and truncation errors.
- **Finite elements** reformulate the problem variationally and approximate the solution in a finite-dimensional space using hat functions.
- Both methods in 1D with uniform grids yield tridiagonal linear systems, which are easy to solve.
- The finite element framework extends naturally to variable coefficients, irregular meshes, and higher dimensions, whereas this is less straightforward for finite differences.

Abstract Hilbert-space framework We now place the previous construction into a general framework. Let $D \subset \mathbb{R}^d$ be an open, bounded set with smooth boundary and $V = V(D)$ a real Hilbert space. Let $\mathcal{L} : V \to V'$ be a second-order elliptic differential operator, e.g. $-\Delta : H_0^1(D) \to H^{-1}(D)$, and let $f \in V'$.

Problem 3.7 (Strong formulation)**.** Find $u \in V$ such that

$$\mathcal{L}u = f \quad \text{in } V'.$$

We denote by $(\cdot,\cdot)_V$ the inner product on V and by $\|\cdot\|_V$ the induced norm. By $\langle\cdot,\cdot\rangle$ we denote the duality pairing between V' and V. We define the bilinear form $a(\cdot,\cdot) : V \times V \to \mathbb{R}$ and linear functional $f(\cdot) : V \to \mathbb{R}$

$$a(u,v) = \langle \mathcal{L}u, v\rangle \quad \forall u,v \in V, \qquad f(v) = \langle f, v\rangle \quad \forall v \in V. \tag{3.11}$$

Problem 3.8 (Variational formulation)**.** Find $u \in V$ such that

$$a(u,v) = f(v) \quad \forall v \in V.$$

We assume that $a(\cdot,\cdot)$ and $f(\cdot)$ are continuous on $V \times V$ and V, i.e., there exist constants $C > 0$ and $C' > 0$ such that

$$|a(u,v)| \le C\|u\|_V\|v\|_V \quad \forall u,v \in V, \qquad |f(v)| \le C'\|v\|_V \quad \forall v \in V. \tag{3.12}$$

We also assume that $a(\cdot,\cdot)$ is coercive on V, i.e., there exists a constant $\alpha > 0$ such that

$$a(u,u) \ge \alpha\|u\|_V^2 \quad \forall u \in V. \tag{3.13}$$

In this case, we have uniqueness to the variational problem.

Theorem 3.9 (Lax–Milgram)**.** *Under the continuity and coercivity assumptions* (3.12)–(3.13)*, Problem 3.8 has a unique solution $u \in V$. Moreover,*

$$\|u\|_V \le \frac{1}{\alpha}\|f\|_{V'}.$$

Proof

Let $f \in V'$ and define the linear operator $A : V \to V'$ by

$$\langle Au, v\rangle = a(u,v) \qquad \forall u,v \in V.$$

By the continuity assumption (3.12), A is bounded:

$$\|Au\|_{V'} = \sup_{\|v\|_V=1} |a(u,v)| \le C\|u\|_V.$$

By coercivity (3.13), we have

$$\langle Au, u\rangle = a(u,u) \ge \alpha\|u\|_V^2,$$

which implies that A is injective and that

$$\|u\|_V \le \frac{1}{\alpha}\|Au\|_{V'}.$$

Thus A is bijective from V onto V' (closed range plus injectivity). Therefore, for every $f \in V'$, there exists a unique $u \in V$ such that

$$Au = f,$$

i.e., $a(u,v) = f(v)\,\forall v \in V$. Finally, using the bound above,

$$\|u\|_V \le \frac{1}{\alpha}\|f\|_{V'}.$$

This completes the proof.

We discretize V by $\{V_h\}_{h>0}$, a sequence of nested, dense finite-dimensional subspaces of V as $h \to 0$.

Problem 3.10 (Galerkin approximation)**.** Find $u_h \in V_h$ such that

$$a(u_h, v_h) = f(v_h) \quad \forall v_h \in V_h.$$

If $a(\cdot,\cdot)$ is continuous and coercive on V, then its restriction to $V_h \times V_h$ is also continuous and coercive, with constants $C_h \leq C$ and $\alpha_h \geq \alpha$, ensuring uniqueness and convergence of the Galerkin approximation.

Theorem 3.11 (Céa's lemma)**.** *Under assumptions* (3.12)–(3.13)*, there exists a unique solution* $u_h \in V_h$ *of Problem 3.10 and*

$$\|u - u_h\|_V \leq \frac{C}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V \xrightarrow[h\to 0]{} 0.$$

Thus, the Galerkin solution is quasi-optimal: it is, up to C/α*, as good as the best approximation in* V_h*.*

Proof

By the definition of u and u_h,

$$a(u, v) = f(v) \quad \forall v \in V, \qquad a(u_h, v_h) = f(v_h) \quad \forall v_h \in V_h.$$

Since $V_h \subset V$, it follows by subtraction that

$$a(u - u_h, v_h) = 0 \quad \forall v_h \in V_h.$$

In particular, since $v_h - u_h \in V_h$,

$$a(u - u_h,\, v_h - u_h) = 0.$$

Moreover, by coercivity and continuity of a,

$$\begin{aligned}\alpha\|u - u_h\|_V^2 &\leq a(u - u_h, u - u_h) = a(u - u_h, u - v_h) + a(u - u_h, v_h - u_h),\\ &\leq C\|u - u_h\|_V \|u - v_h\|_V.\end{aligned}$$

After dividing by $\|u - u_h\|_V$, we obtain

$$\alpha\|u - u_h\|_V \leq C\|u - v_h\|_V,$$

and the assertion is established.

Let $\{\phi_i\}_{i=1}^m$ be a basis of V_h, and let $U \in \mathbb{R}^m$ denote the coefficient vector of u_h. Define

$$A_{ij} = a(\phi_j, \phi_i), \qquad F_i = f(\phi_i), \qquad 1 \leq i, j \leq m, \tag{3.14}$$

so that the Galerkin approximation leads to the linear system:

Problem 3.12 (Galerkin linear system)**.** Find $U \in \mathbb{R}^m$ such that

$$AU = F.$$

Under continuity and coercivity assumptions on $a(\cdot,\cdot)$, the matrix A is positive definite, guaranteeing a unique solution to Problem 3.12. This sparse system can be solved efficiently by direct methods in 1D/2D (e.g., LU factorization) or iterative methods in 3D (e.g., GMRES).

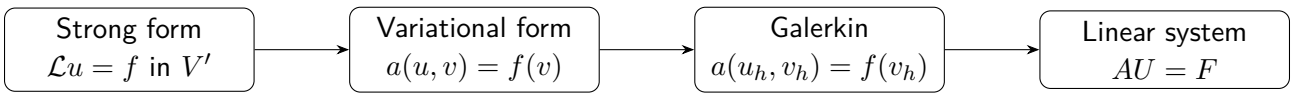

Figure 3: From the strong formulation to the linear system via variational and Galerkin formulations.

Symmetric case and energy minimization If $a(\cdot,\cdot)$ is symmetric, then u solves Problem 3.8 if and only if it minimizes the quadratic functional

$$J(v) = \frac{1}{2}a(v,v) - f(v) \tag{3.15}$$

over V. In this case, the matrix A is symmetric and has eigenvalues $\lambda_j(A) > 0$. One can then use, for instance, Cholesky factorization (direct method) or the conjugate gradient method (iterative).

Approximate forms and Strang's lemma In practice, we often have to approximate the bilinear and linear forms numerically, e.g., by numerical quadrature. Suppose that we can only compute $a_h(\cdot,\cdot)$ and $f_h(\cdot)$, and consider the discrete problem

$$a_h(u_h, v_h) = f_h(v_h) \quad \forall v_h \in V_h,$$

where $a_h(\cdot,\cdot)$ are uniformly C-continuous and α-coercive on $V_h \times V_h$, and $f_h(\cdot)$ are continuous. Then we have Strang's lemma, which separates the error into approximation and consistency parts:

$$\underbrace{\|u-u_h\|_V}_{\textit{solution error}} \le c\left\{ \inf_{v_h\in V_h} \left(\underbrace{\|u-v_h\|_V}_{\textit{approx. error}} + \underbrace{\sup_{w_h\in V_h} \frac{|a(v_h,w_h)-a_h(v_h,w_h)|}{\|w_h\|_V}}_{\textit{consistency error}} \right) + \underbrace{\sup_{w_h\in V_h} \frac{|f(w_h)-f_h(w_h)|}{\|w_h\|_V}}_{\textit{consistency error}} \right\}.$$

This yields the general principle **stability/coercivity** + **consistency** $\Rightarrow$ **convergence**. One may further add a *numerical error* term accounting for the approximate solution of the linear system $AU = F$; in the language of Chapter 1, this is analogous to an *optimization error*.

Proof

Let $v_h \in V_h$. For convenience, set $u_h - v_h = w_h$. Then by uniform coercivity of a_h, we have

$$\begin{aligned}
\alpha\,\|u_h - v_h\|_V^2 &\le a_h(u_h - v_h,\, u_h - v_h) = a_h(u_h - v_h, w_h),\\
&= a(u - v_h, w_h) + [a(v_h,w_h) - a_h(v_h,w_h)] + [a_h(u_h,w_h) - a(u,w_h)],\\
&= a(u - v_h, w_h) + [a(v_h,w_h) - a_h(v_h,w_h)] - [f_h(w_h) - f(w_h)].
\end{aligned}$$

Dividing through by $\|u_h - v_h\|_V = \|w_h\|_V$ and using the continuity of a, we get

$$\|u_h - v_h\|_V \le \frac{C}{\alpha}\|u - v_h\|_V + \frac{1}{\alpha}\frac{|\,a(v_h,w_h) - a_h(v_h,w_h)\,|}{\|w_h\|_V} + \frac{1}{\alpha}\frac{|\,f(w_h) - f_h(w_h)\,|}{\|w_h\|_V}.$$

Since v_h is an arbitrary element of V_h, the assertion follows from the triangle inequality

$$\|u - u_h\|_V \le \|u - v_h\|_V + \|u_h - v_h\|_V.$$

Back to 1D Poisson In our original example, we take

$$V = \{v \in H^1(-1,1) : v(-1) = v(1) = 0\},$$

and the bilinear form $a : V \times V \to \mathbb{R}$ is

$$a(u,v) = \int_{-1}^{1} u'(x)\, v'(x)\, dx.$$

We have continuity for all $u, v \in V$:

$$|a(u,v)| \le \|u'\|_{L^2}\, \|v'\|_{L^2} \le \|u\|_{H^1}\, \|v\|_{H^1}. \tag{3.16}$$

Moreover, by the Poincaré inequality, there exists $C > 0$ such that $\|u\|_{L^2} \le C\|u'\|_{L^2}$ for all $u \in V$, hence

$$a(u,u) = \|u'\|_{L^2}^2 \ge \frac{1}{1+C^2}\,\|u\|_{H^1}^2, \tag{3.17}$$

which shows coercivity. The approximation space is precisely $V_h = \operatorname{span}\{\phi_1, \ldots, \phi_m\} \subset V$, where $\{\phi_j\}$ are the hat functions defined above. The form $a(\cdot,\cdot)$ is symmetric and coercive on V, hence A is SPD.

Finally, we recall that $H_0^1(-1,1)$ is a closed subspace of $H^1(-1,1)$ and therefore itself a Hilbert space for the H^1-norm, which fits perfectly into the abstract framework described above.

4 Discretization of the 1D heat equation

We consider the heat equation on $(0,T)\times(-1,1)$

$$u_t = u_{xx}, \qquad u(t,\pm 1)=0, \qquad u(0,x)=u_0(x).$$

We present two standard discretization schemes: (i) finite differences in space + time stepping, analyzed by von Neumann; (ii) finite elements in space + time stepping, analyzed by eigenmodes of $M^{-1}A$.

4.1 Finite differences

Space-time discretization Let $x_i = -1+ih$ with $h = 2/(m+1)$ and $i = 0,\dots,m+1$, and let $t_n = n\Delta t$ with $n = 0,\dots,N$. We denote by $u_i^n \approx u(t_n,x_i)$ the grid values, with Dirichlet boundary values $u_0^n = u_{m+1}^n = 0$. We use the centered second-difference operator

$$(D_{xx}^h u^n)_i := \frac{u_{i-1}^n - 2u_i^n + u_{i+1}^n}{h^2}, \qquad i = 1,\dots,m.$$

Definition 4.1 (θ-scheme)**.** Discretizing $u_t = u_{xx}$ by the centered second-difference operator in space and a θ-scheme in time gives, for $i = 1,\dots,m$,

$$\frac{u_i^{n+1}-u_i^n}{\Delta t} = (1-\theta)\,(D_{xx}^h u^n)_i + \theta\,(D_{xx}^h u^{n+1})_i, \qquad 0\le\theta\le 1. \tag{4.1}$$

The classical schemes are recovered by

$$\theta = 0 \text{ (forward Euler)}, \qquad \theta = 1 \text{ (backward Euler)}, \qquad \theta = \tfrac{1}{2} \text{ (Crank–Nicolson)}.$$

Two notions will be central.

Definition 4.2 (Space–time accuracy and truncation error)**.** Let $u(t,x)$ be a sufficiently smooth solution of the heat equation. Substituting the exact values $u(t_n,x_i)$ into Eq. (4.1) produces a residual

$$\tau_i^{n+1} := \frac{u(t_{n+1},x_i)-u(t_n,x_i)}{\Delta t} - (1-\theta)(D_{xx}^h u(t_n,\cdot))_i - \theta(D_{xx}^h u(t_{n+1},\cdot))_i,$$

called the *local truncation error*. The scheme is said to be consistent if

$$\max_{i,n}|\tau_i^{n+1}| \longrightarrow 0 \qquad \text{as } \Delta t\to 0 \text{ and } h\to 0.$$

Its order in time and space is determined by how fast τ_i^{n+1} tends to zero with respect to Δt and h.

Von Neumann (Fourier) stability analysis We begin by specifying the notion of stability used throughout.

Definition 4.3 (Stability)**.** A fully discrete numerical scheme is said to be *stable* if errors arising from perturbations in the discrete data—such as initial conditions, boundary conditions, forcing terms, roundoff errors—remain uniformly bounded as the number of time steps $N\to\infty$.

Fourier (von Neumann) stability analysis Von Neumann stability analysis applies only to linear schemes with constant coefficients and is carried out in an idealized setting assuming *periodic solutions*. We therefore consider periodic solutions on the interval $(-1,1)$, extended periodically with period 2.

Expand the discrete solution in Fourier modes,

$$u_i^n = \sum_{k\in\mathbb{Z}} \hat{u}^{\,n}(k)\, e^{\mathrm{i}k\pi x_i}.$$

Each mode evolves independently under the θ-scheme:

$$\hat{u}^{\,n+1}(k) = g_\theta(k)\,\hat{u}^{\,n}(k),$$

where $g_\theta(k)$ is the *amplification factor*. The scheme is stable if

$$|g_\theta(k)|\le 1 \qquad \text{for all } k\in\mathbb{Z}.$$

To compute the amplification factor, consider a single Fourier mode

$$u_i^n = \hat{u}^n e^{\mathrm{i}k\pi x_i}.$$

Using the shift property,

$$e^{\mathrm{i}k\pi x_{i\pm1}} = e^{\pm \mathrm{i}k\pi h}\, e^{\mathrm{i}k\pi x_i},$$

the centered second-difference operator gives

$$(D_{xx}^h u^n)_i = \frac{e^{-\mathrm{i}k\pi h} - 2 + e^{\mathrm{i}k\pi h}}{h^2}\, u_i^n = -\frac{4}{h^2}\sin^2\Big(\frac{k\pi h}{2}\Big) u_i^n.$$

Substituting into the θ-scheme (4.1) yields

$$\hat{u}^{n+1} = g_\theta(k)\,\hat{u}^n, \quad g_\theta(k) = \frac{1 - 4(1-\theta)\mu\sin^2(\frac{k\pi h}{2})}{1 + 4\theta\mu\sin^2(\frac{k\pi h}{2})}, \quad \mu := \frac{\Delta t}{h^2}.$$

The stability for the three main schemes is given in the proposition below.

Proposition 4.4 (Stability of θ-scheme)**.** *Let* $s := \sin^2(\frac{kh\pi}{2}) \in [0,1]$.

- ***Forward Euler*** *($\theta = 0$):*

$$g_0(k) = 1 - 4\mu s, \qquad |g_0(k)| \le 1 \iff \mu \le \frac{1}{2},$$

  i.e., the time step must satisfy $\Delta t \le h^2/2$.

- ***Backward Euler*** *($\theta = 1$):*

$$g_1(k) = \frac{1}{1 + 4\mu s}, \qquad |g_1(k)| \le 1,$$

  so the scheme is unconditionally stable.

- ***Crank–Nicolson*** *($\theta = \frac{1}{2}$):*

$$g_{1/2}(k) = \frac{1 - 2\mu s}{1 + 2\mu s}, \qquad |g_{1/2}(k)| \le 1,$$

  so the scheme is unconditionally stable.

The stability requirement $\Delta t \le h^2/2$ is called the CFL condition, named after Courant, Friedrichs, and Lewy.

4.2 Finite elements

We discretize the heat equation in space by finite elements as before, which leads to a linear ODE system in time. Time is then discretized by the same θ-scheme as in the finite difference setting.

Semi-discretization in space Find $u : [0,T] \to H_0^1(-1,1)$ such that

$$\int_{-1}^{1} u_t v\,dx + \int_{-1}^{1} u_x v_x\,dx = 0 \qquad \forall v \in H_0^1([-1,1]),$$

with $u(0,\cdot) = u_0$. Let $V_h = \operatorname{span}\{\phi_1, \dots, \phi_m\} \subset H_0^1(-1,1)$ be the previous hat-function space and seek

$$u_h(t,x) = \sum_{j=1}^{m} u_j(t)\phi_j(x).$$

Testing with ϕ_i gives the matrix ODE

$$MU'(t) + AU(t) = 0, \qquad U(0) = U^0, \tag{4.2}$$

with coefficient vector $U(t) = (u_1(t), \dots, u_m(t))^\top$ and

$$M_{ij} = \int_{-1}^{1} \phi_j\phi_i\,dx, \qquad A_{ij} = \int_{-1}^{1} \phi_j'\phi_i'\,dx.$$

The stiffness matrix A is the same as for Poisson; M is called the mass matrix. Both M and A are symmetric positive definite. For piecewise linear elements on a uniform mesh one can show that

$$M = \frac{h}{3}\begin{pmatrix} 2 & 0.5 & & & \\ 0.5 & 2 & 0.5 & & \\ & \ddots & \ddots & \ddots & \\ & & 0.5 & 2 & 0.5 \\ & & & 0.5 & 2 \end{pmatrix}, \qquad A = \frac{1}{h}\begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}.$$

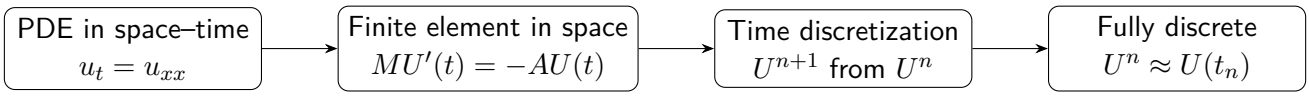

Figure 4: Pipeline: spatial finite elements yield an ODE system, then discretized in time.

Time discretization Let $t_n = n\Delta t$ and $U^n \approx U(t_n)$. Applying the θ-scheme to Eq. (4.2) yields

$$M\frac{U^{n+1} - U^n}{\Delta t} + A((1-\theta)U^n + \theta U^{n+1}) = 0, \qquad 0 \le \theta \le 1, \tag{4.3}$$

Eigenvalue stability analysis The fully discrete θ-scheme reads

$$U^{n+1} = R_\theta(\Delta t)\, U^n, \qquad R_\theta(\Delta t) = (M + \theta\Delta t A)^{-1}(M - (1-\theta)\Delta t A).$$

Since the scheme is linear, stability is equivalent to $\rho(R_\theta(\Delta t)) \le 1$, where $\rho(\cdot)$ denotes the spectral radius.

Proposition 4.5 (Stability of the θ-scheme)**.**

- ***Forward Euler*** *($\theta = 0$): stable iff $\Delta t \le h^2/6$ (CFL condition).*
- ***Backward Euler*** *($\theta = 1$): unconditionally stable.*
- ***Crank–Nicolson*** *($\theta = 1/2$): unconditionally stable.*

Proof

We prove the result for Forward Euler ($\theta = 0$); proofs for the other schemes are similar. In this case,

$$R_0(\Delta t) = I - \Delta t M^{-1} A.$$

Since $M^{-1}A$ is diagonalizable with real positive eigenvalues $\lambda_j(M^{-1}A) > 0$, the eigenvalues of R_0 are

$$\lambda_j(R_0) = 1 - \Delta t \lambda_j(M^{-1}A).$$

Stability requires $|\lambda_j(R_0)| \le 1$ for all j, i.e.,

$$0 \le \Delta t \lambda_j(M^{-1}A) \le 2 \quad \forall j,$$

so the CFL condition is

$$\Delta t \le \frac{2}{\lambda_{\max}(M^{-1}A)}.$$

For standard 1D finite elements on a uniform mesh,

$$\lambda_j(M) = \frac{h}{3}\Big(1 + 2\cos^2\frac{j\pi h}{4}\Big), \quad \lambda_j(A) = \frac{4}{h}\sin^2\frac{j\pi h}{2}, \qquad j = 1, \dots, m,$$

and since M and A are diagonalizable in the same basis, the eigenvalues of $M^{-1}A$ are

$$\lambda_j(M^{-1}A) = \frac{\lambda_j(A)}{\lambda_j(M)} = \frac{12}{h^2}\frac{\sin^2(\frac{j\pi h}{2})}{1 + 2\cos^2(\frac{j\pi h}{4})}, \quad j = 1, \dots, m.$$

Thus the largest eigenvalue satisfies $\lambda_{\max}(M^{-1}A) \le 12/h^2$, giving the CFL restriction $\Delta t \le h^2/6$.

5 Exercises

Exercise 5.1. For the 1D heat equation $u_t = u_{xx}$, the finite element semi-discretization gives

$$MU'(t) = -AU(t).$$

(a) Compute the mass matrix entries

$$M_{ij} = \int_{-1}^{1} \phi_i(x)\phi_j(x)\,dx,$$

and show that for a uniform mesh,

$$M = \frac{h}{6}\begin{pmatrix} 4 & 1 & & & \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 4 & 1 \\ & & & 1 & 4 \end{pmatrix}.$$

(b) Explain briefly the physical meaning of M and A in the semi-discrete ODE.

Exercise 5.2. Consider the Poisson problem on $[-1, 1]$ with $V = H_0^1(-1, 1)$ and

$$a(u, v) = \int_{-1}^{1} u'(x)v'(x)\,dx.$$

(a) Prove that $a(\cdot, \cdot)$ is an inner product on V.

(b) Prove the equivalence of the energy norm

$$\|u\|_a := a(u, u)^{1/2}$$

with the standard H^1-norm on V using the Poincaré inequality.

(c) Determine explicit constants $c_1, c_2 > 0$ such that

$$c_1\|u\|_{H^1} \le \|u\|_a \le c_2\|u\|_{H^1}.$$

Exercise 5.3. Let $V_h \subset V$ be the continuous, piecewise linear finite element space.

(a) Show that the Galerkin solution u_h satisfies

$$a(u - u_h, v_h) = 0 \qquad \forall v_h \in V_h.$$

(b) Using part (a), prove the orthogonality-based estimate

$$\|u - u_h\|_a = \inf_{v_h \in V_h} \|u - v_h\|_a.$$

(c) Conclude Céa's lemma directly from the norm equivalence.

Exercise 5.4. Let $I_h : H_0^1(-1, 1) \to V_h$ denote the nodal interpolation operator.

(a) Show that for any $u \in H^2(-1, 1)$, the interpolation error satisfies

$$\|u - I_h u\|_{L^2} \le Ch^2\|u''\|_{L^2}.$$

(b) Show that in the energy norm,

$$\|u - I_h u\|_a \le Ch\|u''\|_{L^2}.$$

(c) Explain why part (b), combined with Céa's lemma, implies first-order convergence for u_h.

6 Suggested reading

To learn more about nonlinear ODEs and dynamics, we recommend checking out APM 41012, along with [11, 15]. For those interested in ODE simulations, APM 41012 is a great place to start, and [7] is a helpful and exhaustive resource. For learning about PDE theory, we suggest the course FMA 42032, along with important texts in [4, 5, 9, 14]. For finite difference methods, APM 41012 gives a good introduction, and the classics [8, 10, 12] provide more details. For finite element methods, you can check out APM 42031 for theory, and APM 52184 for practical implementation. You can also refer to [1, 2, 3]. For spectral methods, see [6, 13].

Table 1: *TL;DR table of 15 favorite references.*

	Courses @ X	Books
ODE theory	APM 41012	[11, 15]
ODE simulation	APM 41012	[7]
PDE theory	FMA 42032	[4, 5, 9, 14]
Finite differences	APM 41012	[8, 10, 12]
Finite elements	APM 42031 & APM 52184	[1, 2, 3]
Spectral methods	–	[6, 13]

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